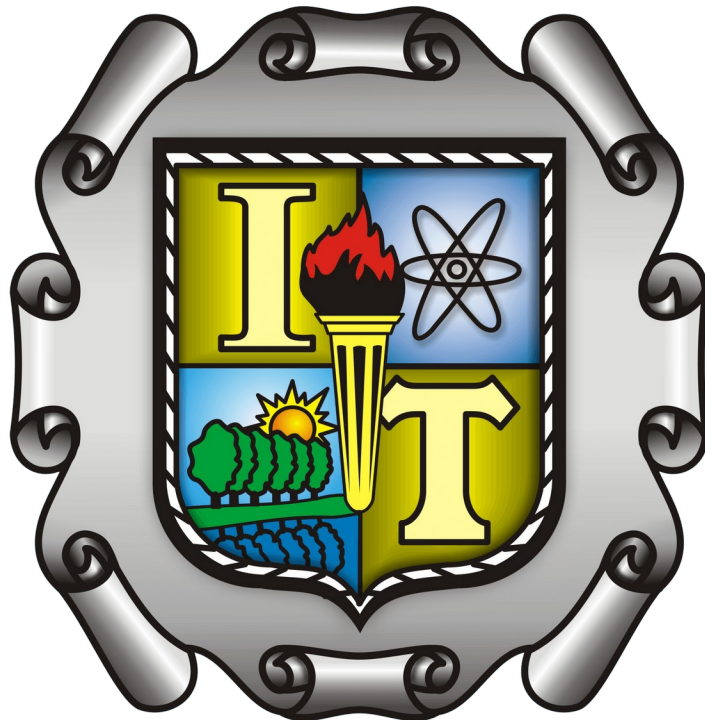


Tecnologico Nacional de Mexico

Instituto Tecnológico de Saltillo

Computo en la Nube

Seguridad (hardening) en servidor web Apache 2



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```

root@CN:~# sudo apt install apache2 -y
Upgrading:
 apache2 apache2-bin apache2-data apache2-utils

Summary:
 Upgrading: 4, Installing: 0, Removing: 0, Not Upgrading: 81
 Download size: 2 011 kB
 Space needed: 0 B / 14.4 GB available

Des:1 http://deb.debian.org/debian trixie/main amd64 apache2 amd64 2.4.66-1~deb13u2 [225 kB]
Des:2 http://deb.debian.org/debian trixie/main amd64 apache2-bin amd64 2.4.66-1~deb13u2 [1 409 kB]
Des:3 http://deb.debian.org/debian trixie/main amd64 apache2-data all 2.4.66-1~deb13u2 [160 kB]
Des:4 http://deb.debian.org/debian trixie/main amd64 apache2-utils amd64 2.4.66-1~deb13u2 [216 kB]
Descargados 2 011 kB en 12s (169 kB/s)
apt-listchanges: Leyendo lista de cambios...
(Leyendo la base de datos ... 78137 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../apache2_2.4.66-1~deb13u2_amd64.deb ...
Desempaquetando apache2 (2.4.66-1~deb13u2) sobre (2.4.66-1~deb13u1) ...
Preparando para desempaquetar .../apache2-bin_2.4.66-1~deb13u2_amd64.deb ...
Desempaquetando apache2-bin (2.4.66-1~deb13u2) sobre (2.4.66-1~deb13u1) ...
Preparando para desempaquetar .../apache2-data_2.4.66-1~deb13u2_all.deb ...
Desempaquetando apache2-data (2.4.66-1~deb13u2) sobre (2.4.66-1~deb13u1) ...
Preparando para desempaquetar .../apache2-utils_2.4.66-1~deb13u2_amd64.deb ...
Desempaquetando apache2-utils (2.4.66-1~deb13u2) sobre (2.4.66-1~deb13u1) ...
Configurando apache2-bin (2.4.66-1~deb13u2) ...
Configurando apache2-data (2.4.66-1~deb13u2) ...
Configurando apache2-utils (2.4.66-1~deb13u2) ...
Configurando apache2 (2.4.66-1~deb13u2) ...
apache-htcacheclean.service is a disabled or a static unit not running, not starting it.
Procesando disparadores para man-db (2.13.1-1) ...
root@CN:~#

```



Apache2 Default Page

Ubuntu

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is **fully documented in `/usr/share/doc/apache2/README.Debian.gz`**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:

```

/etc/apache2/
|-- apache2.conf
|   |-- ports.conf
|-- mods-enabled
|   |-- *.load
|   |-- *.conf
|-- conf-enabled
|   |-- *.conf
|-- sites-enabled
|   |-- *.conf

```

- `apache2.conf` is the main configuration file. It puts the pieces together by including all remaining configuration files when starting up the web server.

```
GNU nano 8.4 /etc/apache2/conf-available/security.conf ^
# Changing the following options will not really affect the security of the
# server, but might make attacks slightly more difficult in some cases.
#
# ServerTokens
# This directive configures what you return as the Server HTTP response
# Header. The default is 'Full' which sends information about the OS-Type
# and compiled in modules.
# Set to one of: Full | OS | Minimal | Minor | Major | Prod
# where Full conveys the most information, and Prod the least.
#ServerTokens Minimal
ServerTokens Prod
#ServerTokens Full
#
# Optionally add a line containing the server version and virtual host
# name to server-generated pages (internal error documents, FTP directory
# listings, mod_status and mod_info output etc., but not CGI generated
# documents or custom error documents).
# Set to "EMail" to also include a mailto: link to the ServerAdmin.
# Set to one of: On | Off | EMail
#ServerSignature Off
ServerSignature off
#
# Allow TRACE method
```

```
root@CN:~# sudo a2dismod autoindex status proxy proxy_http
WARNING: The following essential module will be disabled.
This might result in unexpected behavior and should NOT be done
unless you know exactly what you are doing!
autoindex
```

```
To continue type in the phrase 'Yes, do as I say!' or retry by passing '-f': yes
Aborting
root@CN:~# sudo systemctl restart apache2
root@CN:~#
```

```

Des:1 http://deb.debian.org/debian trixie/main amd64 openssl-provider-legacy amd64 3.5.5-1~deb13u1 [311 kB]
Des:2 http://deb.debian.org/debian trixie/main amd64 libssl3t64 amd64 3.5.5-1~deb13u1 [2 442 kB]
Des:3 http://deb.debian.org/debian trixie/main amd64 openssl amd64 3.5.5-1~deb13u1 [1 499 kB]
Descargados 4 252 kB en 2s (1 714 kB/s)
apt-listchanges: Leyendo lista de cambios...
(Leyendo la base de datos ... 78137 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../openssl-provider-legacy_3.5.5-1~deb13u1_amd64.deb ...
Desempaquetando openssl-provider-legacy (3.5.5-1~deb13u1) sobre (3.5.4-1~deb13u2) ...
Configurando openssl-provider-legacy (3.5.5-1~deb13u1) ...
(Leyendo la base de datos ... 78137 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../libssl3t64_3.5.5-1~deb13u1_amd64.deb ...
Desempaquetando libssl3t64:amd64 (3.5.5-1~deb13u1) sobre (3.5.4-1~deb13u2) ...
Configurando libssl3t64:amd64 (3.5.5-1~deb13u1) ...
(Leyendo la base de datos ... 78137 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../openssl_3.5.5-1~deb13u1_amd64.deb ...
Desempaquetando openssl (3.5.5-1~deb13u1) sobre (3.5.4-1~deb13u2) ...
Configurando openssl (3.5.5-1~deb13u1) ...
Procesando disparadores para man-db (2.13.1-1) ...
Procesando disparadores para libc-bin (2.41-12+deb13u1) ...
root@CN:~# sudo a2enmod ssl
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create self-signed certificates.
To activate the new configuration, you need to run:
    systemctl restart apache2
root@CN:~# █

```

←

→

↻

📍 localhost

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Importar marcadores...

🌐 ChatGPT

🦋 DeepSeek - Into the U...

🌐 Facebook

📌 Tablero

🌐 classroom

No se puede conectar

Firefox no puede establecer una conexión con el servidor en localhost.

- El sitio podría estar no disponible temporalmente o demasiado ocupado. Vuelva a intentarlo en unos momentos.
- Si no puede cargar ninguna página, compruebe la conexión de red de su equipo.
- Si su equipo o red están protegidos por un cortafuegos o proxy, asegúrese de que Firefox tiene permiso para acceder a la web.

Reintentar


```
Procesando disparadores para man-db (2.13.1-1) ...
root@CN:~# sudo ufw allow 80
sudo ufw allow 443
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
root@CN:~# sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
root@CN:~# sudo ufw allow ssh
Rule added
Rule added (v6)
root@CN:~#
```

```
root@CN:~# sudo tail -f /var/log/apache2/access.log
192.168.100.7 - - [17/Feb/2026:15:05:35 -0600] "GET /usuarios/repositorio/css/jquery.dataTables.min.css HTTP/1.1" 200
2446 "http://192.168.100.108/usuarios/controladores/controlador.php?gui=usuario" "Mozilla/5.0 (X11; Linux x86_64; rv:1
47.0) Gecko/20100101 Firefox/147.0"
192.168.100.7 - - [17/Feb/2026:15:05:35 -0600] "GET /usuarios/repositorio/css/style.css HTTP/1.1" 200 539 "http://192.
168.100.108/usuarios/controladores/controlador.php?gui=usuario" "Mozilla/5.0 (X11; Linux x86_64; rv:147.0) Gecko/20100
101 Firefox/147.0"
192.168.100.7 - - [17/Feb/2026:15:05:35 -0600] "GET /usuarios/repositorio/datatables/dataTables.bootstrap.css HTTP/1.1
" 200 1365 "http://192.168.100.108/usuarios/controladores/controlador.php?gui=usuario" "Mozilla/5.0 (X11; Linux x86_64
; rv:147.0) Gecko/20100101 Firefox/147.0"
192.168.100.7 - - [17/Feb/2026:15:05:35 -0600] "GET /usuarios/repositorio/datatables/datatable.js HTTP/1.1" 200 730 "h
ttp://192.168.100.108/usuarios/controladores/controlador.php?gui=usuario" "Mozilla/5.0 (X11; Linux x86_64; rv:147.0) G
ecko/20100101 Firefox/147.0"
192.168.100.7 - - [17/Feb/2026:15:05:35 -0600] "GET /usuarios/repositorio/css/bootstrap-3.3.5.min.css HTTP/1.1" 200 20
039 "http://192.168.100.108/usuarios/controladores/controlador.php?gui=usuario" "Mozilla/5.0 (X11; Linux x86_64; rv:14
7.0) Gecko/20100101 Firefox/147.0"
192.168.100.7 - - [17/Feb/2026:15:05:35 -0600] "GET /usuarios/repositorio/js/jquery-3.3.1.js HTTP/1.1" 200 30653 "http
://192.168.100.108/usuarios/controladores/controlador.php?gui=usuario" "Mozilla/5.0 (X11; Linux x86_64; rv:147.0) Geck
o/20100101 Firefox/147.0"
192.168.100.7 - - [17/Feb/2026:15:05:35 -0600] "GET /usuarios/repositorio/js/jquery.dataTables.js HTTP/1.1" 200 113714
"http://192.168.100.108/usuarios/controladores/controlador.php?gui=usuario" "Mozilla/5.0 (X11; Linux x86_64; rv:147.0
) Gecko/20100101 Firefox/147.0"
192.168.100.7 - - [17/Feb/2026:15:05:35 -0600] "GET /usuarios/repositorio/fonts/glyphicons-halflings-regular.woff2 HTT
P/1.1" 200 18316 "http://192.168.100.108/usuarios/repositorio/css/bootstrap-3.3.5.min.css" "Mozilla/5.0 (X11; Linux x8
6_64; rv:147.0) Gecko/20100101 Firefox/147.0"
:::1 - - [17/Feb/2026:15:05:41 -0600] "OPTIONS * HTTP/1.0" 200 126 "-" "Apache/2.4.66 (Debian) (internal dummy connecti
on)"
```

```
root@CN:~# sudo tail -f /var/log/apache2/error.log
[Fri Mar 20 07:37:03.281047 2026] [core:notice] [pid 1047:tid 1047] AH00094: Command line: '/usr/sbin/apache2'
[Fri Mar 20 07:37:27.046243 2026] [mpm_prefork:notice] [pid 1047:tid 1047] AH00170: caught SIGWINCH, shutting down gra
cefully
[Fri Mar 20 07:37:27.184523 2026] [mpm_prefork:notice] [pid 1076:tid 1076] AH00163: Apache/2.4.66 (Debian) configured
-- resuming normal operations
[Fri Mar 20 07:37:27.184582 2026] [core:notice] [pid 1076:tid 1076] AH00094: Command line: '/usr/sbin/apache2'
[Fri Mar 20 07:42:28.151716 2026] [mpm_prefork:notice] [pid 1076:tid 1076] AH00171: Graceful restart requested, doing
restart
[Fri Mar 20 07:42:28.212707 2026] [ssl:warn] [pid 1076:tid 1076] AH01882: Init: this version of mod_ssl was compiled a
gainst a newer library (OpenSSL 3.5.5 27 Jan 2026 (OpenSSL 3.5.4 30 Sep 2025), version currently loaded is 0x30500040)
- may result in undefined or erroneous behavior
[Fri Mar 20 07:42:28.216556 2026] [ssl:warn] [pid 1076:tid 1076] AH01906: CN.myquest.virtualbox.org:443:0 server certi
ficate is a CA certificate (BasicConstraints: CA == TRUE !?)
[Fri Mar 20 07:42:28.216576 2026] [ssl:warn] [pid 1076:tid 1076] AH01909: CN.myquest.virtualbox.org:443:0 server certi
ficate does NOT include an ID which matches the server name
[Fri Mar 20 07:42:28.216935 2026] [mpm_prefork:notice] [pid 1076:tid 1076] AH00163: Apache/2.4.66 (Debian) OpenSSL/3.5
.4 configured -- resuming normal operations
[Fri Mar 20 07:42:28.216944 2026] [core:notice] [pid 1076:tid 1076] AH00094: Command line: '/usr/sbin/apache2'
```

Conclusiones

En esta práctica aprendí cómo instalar y asegurar un servidor Apache, viendo que no solo es importante que funcione, sino también que esté protegido. Me di cuenta de que cosas como ocultar la versión del servidor, desactivar módulos que no se usan y activar HTTPS ayudan a reducir riesgos de ataques.